

# A Comparative Analysis of PM- and MDI-QKD – Project Outline

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## Abstract

Quantum Key Distribution (QKD) was first proposed in 1984 by Bennett and Brassard through their foundational Prepare-and-Measure (PM) QKD protocol, now commonly known as BB84. The literature surrounding the topic lacks a unified analysis of how BB84 compares to a more recent, security-forward protocol design approach, Measurement-Device-Independence (MDI). We seek to address this research gap by analysing metrics of (i) key rate over distance for varying topologies, (ii) cost of implementation of optimal topology, both economic and energetic, and (iii) scalability of topologies for each protocol.

## 1 Motivation, Aims and Research Contribution

QKD refers to any protocol in which two (or more) parties use quantum communication, together with authenticated classical communication and post-processing, to establish a shared secret cryptographic key whose security is guaranteed by laws of quantum mechanics - that is, information-theoretic security against any eavesdropper allowed by quantum physics. Since Bennett and Brassard's exposition in 1984, [1], numerous variations of QKD have been proposed, each optimised for different practical or security-related considerations.

MDI-QKD was proposed in 2012 by Lo, Curty, and Qi, [2], in the form of a novel protocol removing the need for trust in the detectors upon which protocols to that point relied. By establishing a 3rd-party relay, Charlie, to sit in between Alice and Bob, who entangles the pair's photons and performs Bell state measurements (BSMs), instead of direct photon state readings, the need to trust detectors that is assumed in BB84 is removed. Given the difficulty of the problem to consistently fully classify all detector side-channels, BB84 raises a major security vulnerability in practice. As such, [2] was proposed to improve the security of communication of a secret shared key, and presents an improvement to maximum secure key communication distance.

However, in increasing security of communication, the key rate drops (e.g. [3]) in all comparisons with BB84. Initially, Lo et al. compared their MDI protocol to an entanglement-based (EB) scheme proposed by Ma et al., [4], in 2007. Since then, brief comparisons have been synthesised alongside other protocols in surveys such as [5], and refinements to the original protocol have been proposed, such as [6]. However, no unified, comprehensive comparison exists in the literature to understand how BB84 differs from the state-of-the-art of MDI-QKD with respect to: (i) key rate/distance relationships for varying topologies, (ii) economic cost of implementation of optimal topologies, and (iii) effects on cost and key rates of scaling networks (optimal topologies again) in user count.

The successful completion of this analysis may lead to a clear identification of parameter regimes and network requirements under which our MDI-QKD protocol is preferable to BB84, and vice versa - allowing for industry and research to better identify suitable implementations of QKD for a use case. This would advance our understanding of rate and cost scaling for MDI-QKD, and bring us closer to achieving a "quantum internet", put forward in 2008, [7], by informing and simplifying selection processes for key-sharing protocols.

## 2 Methods and Milestones

We address the research gap by modelling an MDI-QKD protocol specified by the use of:

- **Single Photon Sources** (SPSs), achieved through quantum dots (as in [8] and earlier, [9]),
- **BB84 states with polarisation encoding** by using modulators as in the original protocol,
- **BSMs** at the central relay as previously in MDI-QKD,
- **Low-Density Parity Check** error correction and universal hashing for privacy amplification.

Simulations will be led by mathematical modelling and implemented through the use of QuNetSim, [10], and NetSquid, [11], the latter used to model MDI-QKD networks in [12], both being Python frameworks for quantum network simulation. QuNetSim will be used for 2-node BB84 simulations, given its portability and ease of implementation, and NetSquid will be used to model all MDI-QKD investigated and the scaling of all networks. These simulations will lead us to an understanding of the key rate behaviour for both protocols, as well as the effects of network scaling on this metric. Mathematical models will guide the cost analysis, and we will seek to use a universalised "quantum cost" to compare the two protocols.

We track our progress through the following milestones:

#### M1 Mathematical Modelling Complete

- 1.1 Selection and validation of a universal quantum cost function to standardise economic analysis of protocols,
- 1.2 Validation of modelling on top of which to build network simulations;

#### M2 Simulations Constructed

- 2.1 QuNetSim and NetSquid (first pass) simulations built and tested in Python;

#### M3 Initial Simulation Results

- 3.1 First pass at simulations of 2-node networks complete in QuNetSim for BB84 and NetSquid for MDI,
- 3.2 Analysis of initial results complete and drafted;

#### M4 Simulations Finalised

- 4.1 Final simulations of arbitrarily (to a limit) scaled network user counts complete,
- 4.2 Simulation results collated and analysed;

#### M5 Report Complete

- 5.1 All results and analysis collated into final report.

## 3 Timeline and Risk Assessment

- *November – January 2026*: Completion of **M1**, NetSquid and QuNetSim understood, core references fully reviewed for Literature Review portion, supplementary sources identified and reviews underway;
- *January – February*: Completion of **M2**, supplementary source reviews continue, first draft of Literature Review underway, introductory sections of final report in draft;
- *February – March*: Completion of **M3**, Progress Report submission, report introduction complete, Literature Review drafting to continue and literature review portion of final report to begin;
- *March – April*: Literature Review complete, **M4.1** in progress;
- *April – June*: First drafting of report to begin, Case Study to commence with topic selection and preliminary reviews;
- *June – August*: Completion of **M4**, Case Study report to be completed and presented, final drafting process to begin;
- *August – September*: Completion of **M5**, final report submission and viva.

Key risks to the timeline include: **(i)** high module concentration in the first term (4 taught modules till mid-Dec) taking time away from the project, which is mitigated by the lower module count in the second term (2 modules till late Mar), **(ii)** exam period (May-Jun) interfering with report drafting and final results generation more than accounted for, to be mitigated by re-balancing case study work where appropriate, and **(iii)** high complexity to simulate large networks, which can be mitigated by simplifying noise and error modelling, as well as using high-performance computing techniques.

## 4 Guiding References

Key references include [1], [2], [5] and [6] from the below.

### References

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